MINESWEEPERS
TOWARDS A LANDMINE-FREE WORLD

Minesweepers Competition 2026
13th Edition

TEAM NAME
Category

THE CHALLENGE

The mission and our answer

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The arena demands	Our design response
Rough 20 × 20 m arena: sand, grass, inclines, ditches	**Tracked chassis with 15 cm wide tracks**
Buried 10 cm cubes (≤ 10 cm deep) and 5 cm surface cubes	**Metal detector + camera with on-laptop vision**
Light and siren on every detection	**Light + siren fire on every metal hit**
Automatic mine map on a 19 × 19 grid (A–S, 1–19)	**Kalman localization + automatic mapping**
Collect the surface mines	**Autonomous pickup sequence: arm + electromagnet**
No Wi-Fi and no power provided at the arena	**Onboard access point, 2 × 10 Ah LiPo**

OVERVIEW

Robot at a glance

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80 × 75 × 46 cm
Length × width × height

2 × 250 W
24 V geared track motors

Pi 4 + STM32
High-level + real-time control

2 × 10 Ah
LiPo battery packs

Own Wi-Fi
Onboard access point

ROS 2
Robot and base station

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01

Mechanical Systems & Robot Structure

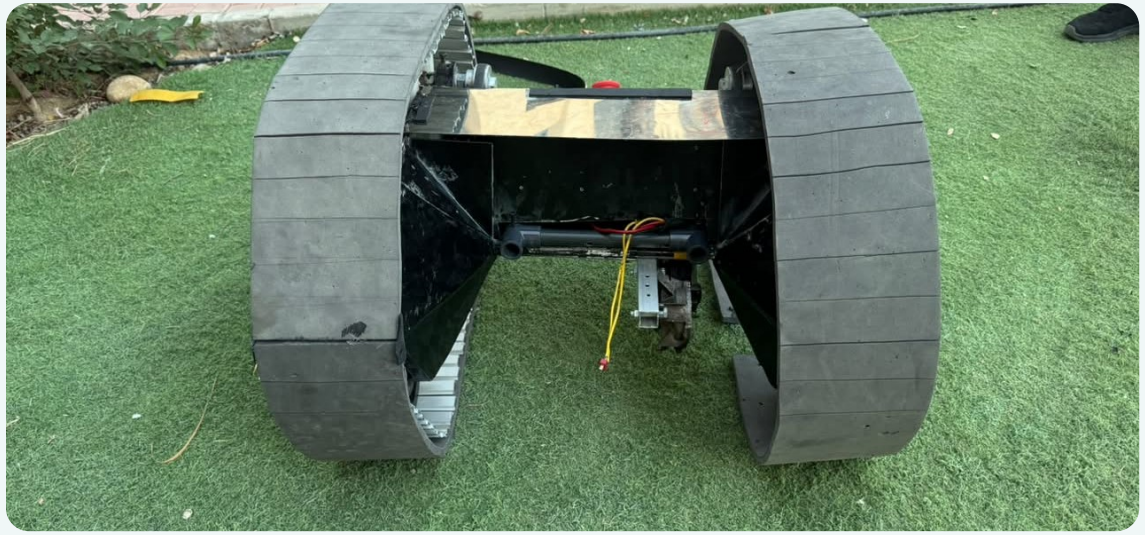
- Body structure
- Why tracks over wheels on sand
- Track construction and drive train
- Arm and autonomous pickup
- Built for sand and heat



MECHANICAL

Body structure

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1 Two track pods

Each pod carries its own motor, chain drive, idlers and road wheels.

2 Central spine

Bridges the pods and carries the electronics, control panel and arm.

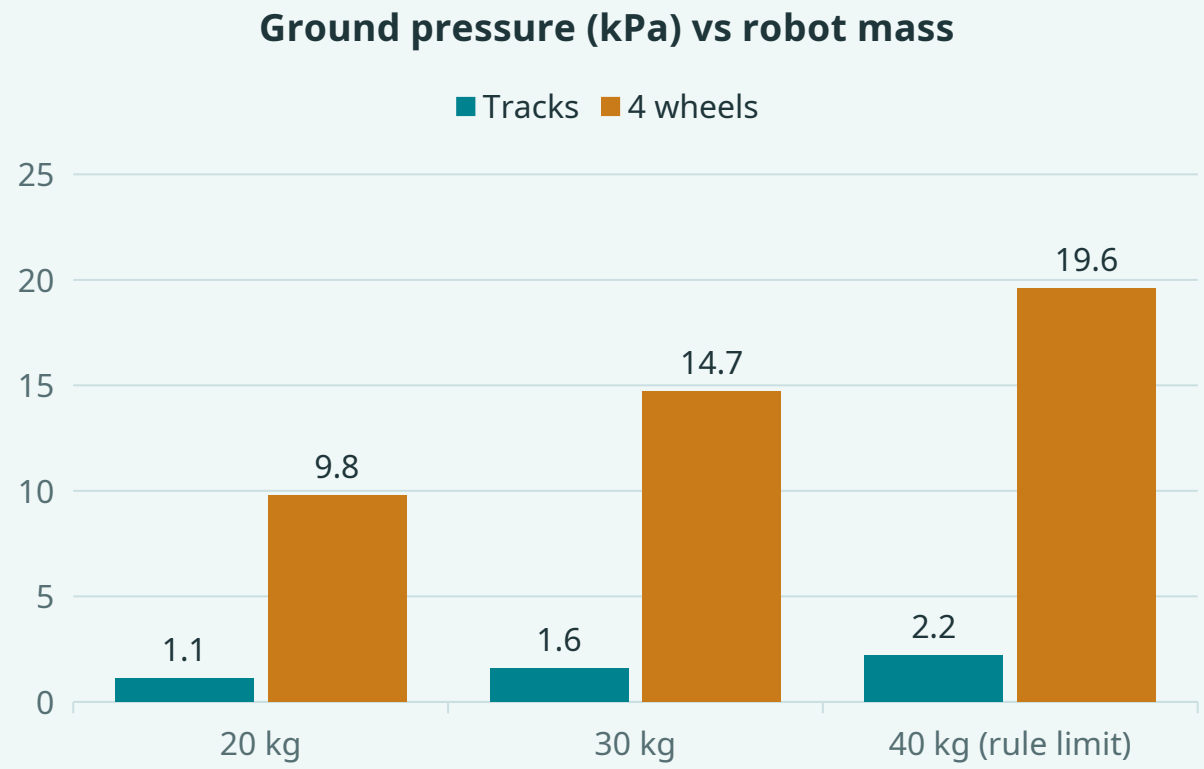
3 Low and wide

80 × 75 × 46 cm footprint keeps the centre of gravity low between the tracks.

MECHANICAL

Why tracks over wheels on sand

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Contact estimates: 2 tracks × 0.60 m × 0.15 m vs 4 wheels × 10 × 5 cm patch.

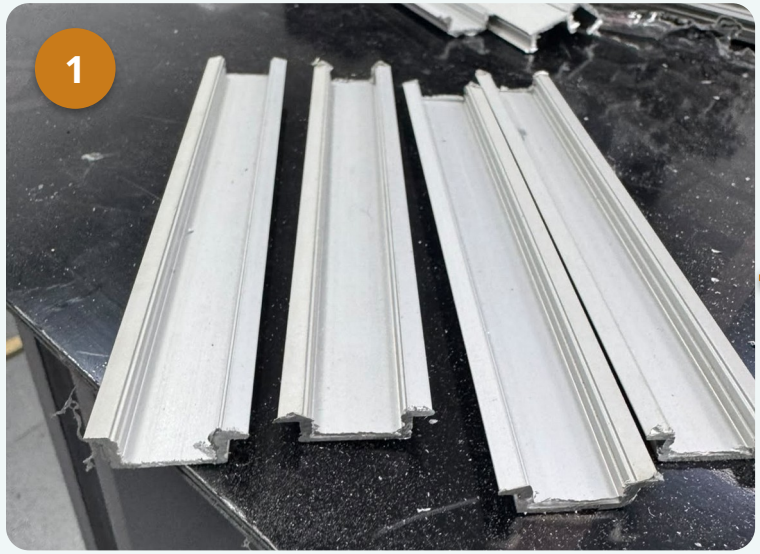
≈ 9× lower ground pressure

- **Floats on loose sand:** the long footprint spreads the load instead of digging in.
- **Traction:** every slat is a grouser biting into sand and grass.
- **Bridges ditches and stones** that would trap a small wheel.
- **Trade-off:** skid-steer turns need more torque, so we use 250 W geared motors.

MECHANICAL

Track construction

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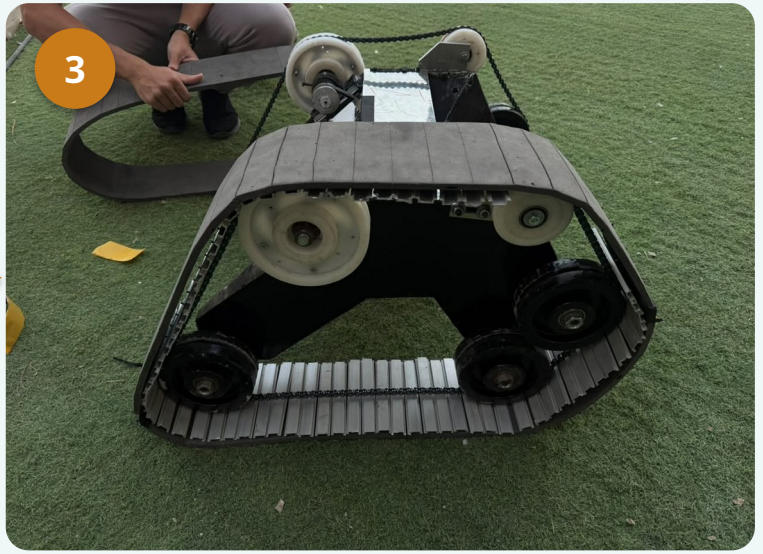
Aluminium slats

Extruded profile with a stepped edge that acts as a grouser.



Chain drive

The chain carries motor torque from the sprocket to every slat.



Rubber grip layer

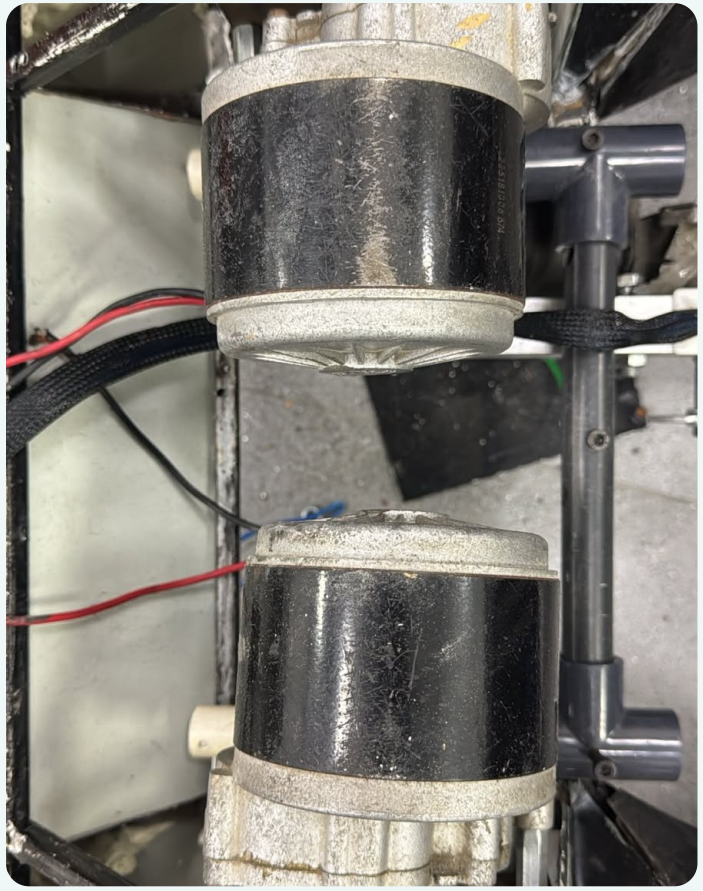
Rubber sheet with cuts so it fits around the slats and grips the sand.

Track loop 195 cm · **width 15 cm** · white idlers on top, road wheels below carry the load

MECHANICAL

Drive train

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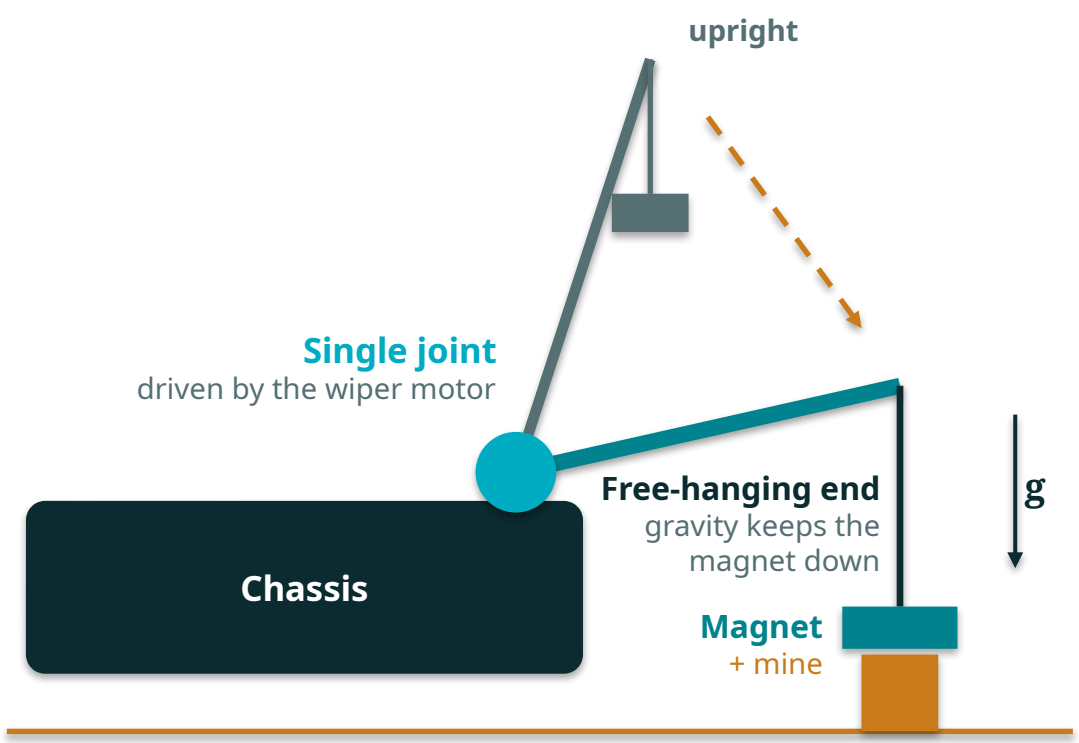
Motor	MY1016Z brushed DC gear motor, 2 units
Origin	Electric wheelchair drive, sourced locally
Supply	24 V DC
Rated power	250 W each, 500 W total
Gearbox	Integrated reduction, high output torque
Current	≈ 10.4 A each at rated output (250 W ÷ 24 V)
Transmission	Chain from gearbox shaft to track sprocket

Skid steering

Same speed on both tracks drives straight; a speed difference turns the robot, down to turning in place. The STM32 mixes the forward and turn commands.

Arm design: one joint, free-hanging end

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1 One degree of freedom

A single rotary joint driven by the wiper motor: one motor, one limit switch, no arm kinematics to solve.

2 Free-weight end

The electromagnet hangs from a free pivot, so gravity keeps it facing the ground at any arm angle, even on a slope.

3 Simple, effective control

Only two positions, lowered and upright, with the upright switch as the reference: easy for the STM32 to control reliably.

Autonomous pickup sequence

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1 Trigger

Operator presses Pick up (J) once. The GUI sends it only if the arm reads upright.

2 Lower

The wiper motor swings the arm down over the surface mine.

3 Grab

The electromagnet is energised and the arm lifts the steel cube.

4 Stow

The magnet releases the mine into the container; the arm returns upright.

Arm motor: WD1160-series wiper motor

12 V DC · aluminium die-cast gearbox · two speeds

Rated 21.5–28 rpm, 5.2–13.2 N·m (peak up to 15 N·m) depending on variant and speed

Runs autonomously

Once triggered, the STM32 runs every step with no operator input. An upright limit switch blocks a new pickup until the arm is back.

Built for sand and heat

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Sand and rough terrain

- Low ground pressure keeps the robot on top of loose sand
- Slat grousers + rubber grip layer for traction on sand and grass
- Long track footprint bridges ditches and stones
- Tested outdoors on sand and grass

Heat

- Electronics sit inside the chassis, shaded from direct sun
- Drive motors cruise well below their 250 W rating
- LiPo temperature checked between runs; spare pack ready

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02

Electronic Components & Control System

- Sensors and microcontrollers
- Why an STM32 for real-time control
- Power architecture and budget
- On-robot and robot-to-station links
- Layered safety stops



ELECTRONICS

Sensors and controllers

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Component	Role	Interface
Raspberry Pi 4	ROS 2, localization, video streaming	Main computer
STM32 Black Pill	Motor PWM, encoders, ADC, arm, magnet, watchdog	UART to Pi
Wheel encoders	Track odometry	STM32 timers
MPU-9250 IMU	9-axis: accelerometer, gyro, magnetometer	I2C to Pi
Metal detector	Buried and surface mines	Analog to STM32 ADC
USB camera	Surface-mine vision and operator view	USB to Pi
Motor drivers	2 × 250 W track motors + arm motor	PWM from STM32
Electromagnet + limit switch	Mine pickup, arm-upright interlock	STM32 I/O
Light + siren	Signal every detection	Digital output
Wi-Fi access point	Robot's own network to the base station	Network

ELECTRONICS

Why an STM32 for real-time control

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1 Hardware encoder timers

Quadrature counting in hardware: no lost counts at full track speed.

2 Deterministic timing

A bare-metal control loop with fixed timing for PWM, sampling and the watchdog.

3 12-bit ADC

Samples the metal-detector signal for clean threshold detection.

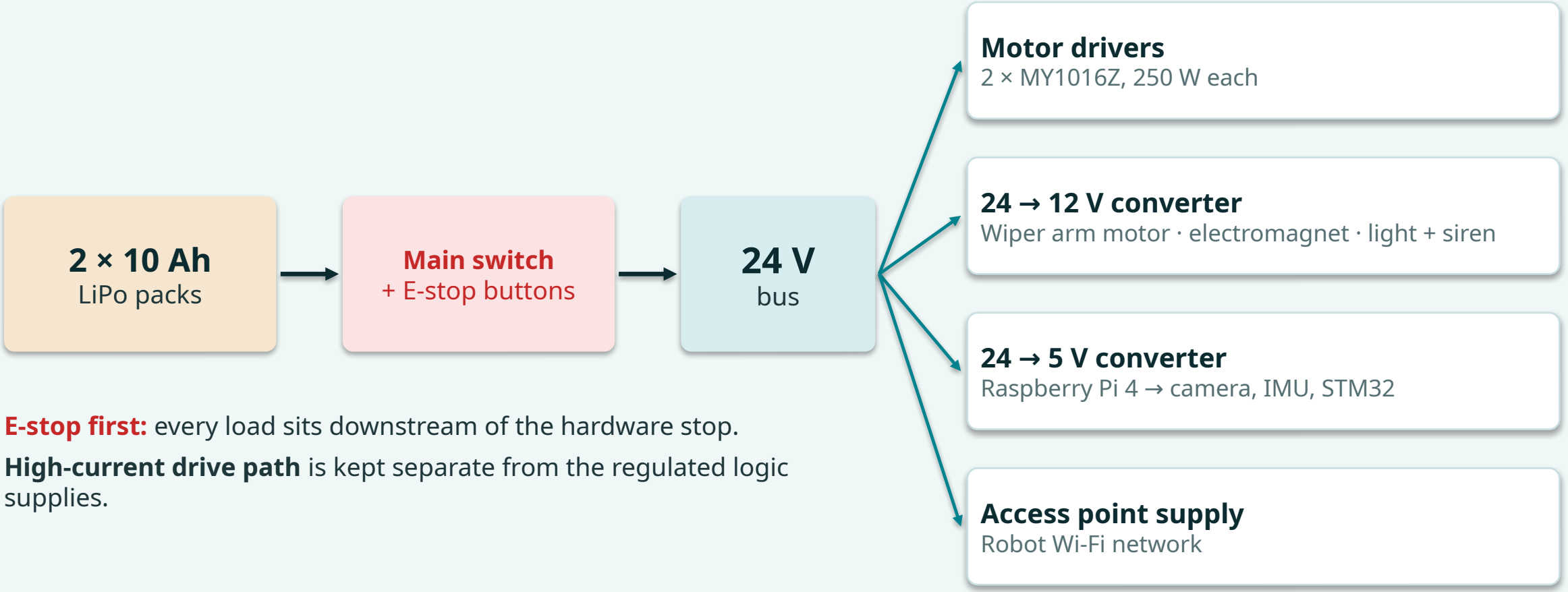
4 Dedicated to control

Networking lives on the Pi and the onboard AP; the MCU only runs real-time I/O.

ELECTRONICS

Power architecture

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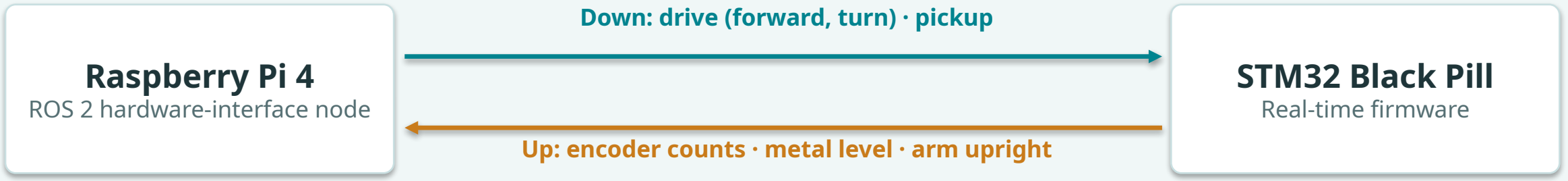


E-stop first: every load sits downstream of the hardware stop.

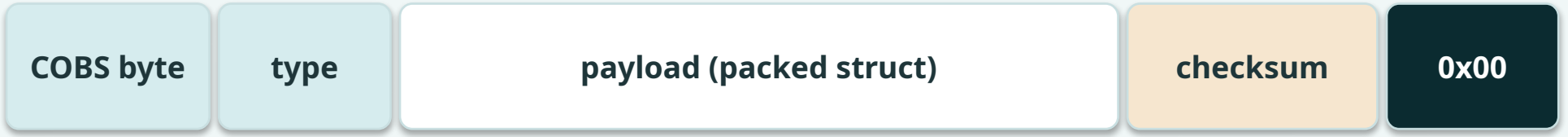
High-current drive path is kept separate from the regulated logic supplies.

Pi to STM32 link

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UART frame



COBS framing

Zero bytes appear only as the delimiter, so the receiver always resynchronises on the next frame.

Checksum

A corrupted frame is dropped, never acted on.

250 ms watchdog

No valid command for 250 ms → the STM32 sets motor PWM to zero.

Robot to control station: our own network

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Robot

Onboard Wi-Fi access point

Raspberry Pi 4
ROS 2 nodes · RTSP server

STM32 + sensors

ROS 2 DDS
drive at 20 Hz · odometry · metal · arm

RTSP video, H.264
kept out of DDS to save bandwidth

Why on the robot? The arena provides no Wi-Fi, so the link is the same in the lab and at the competition.

Base station (outside the fence)

Laptop joins the robot's AP

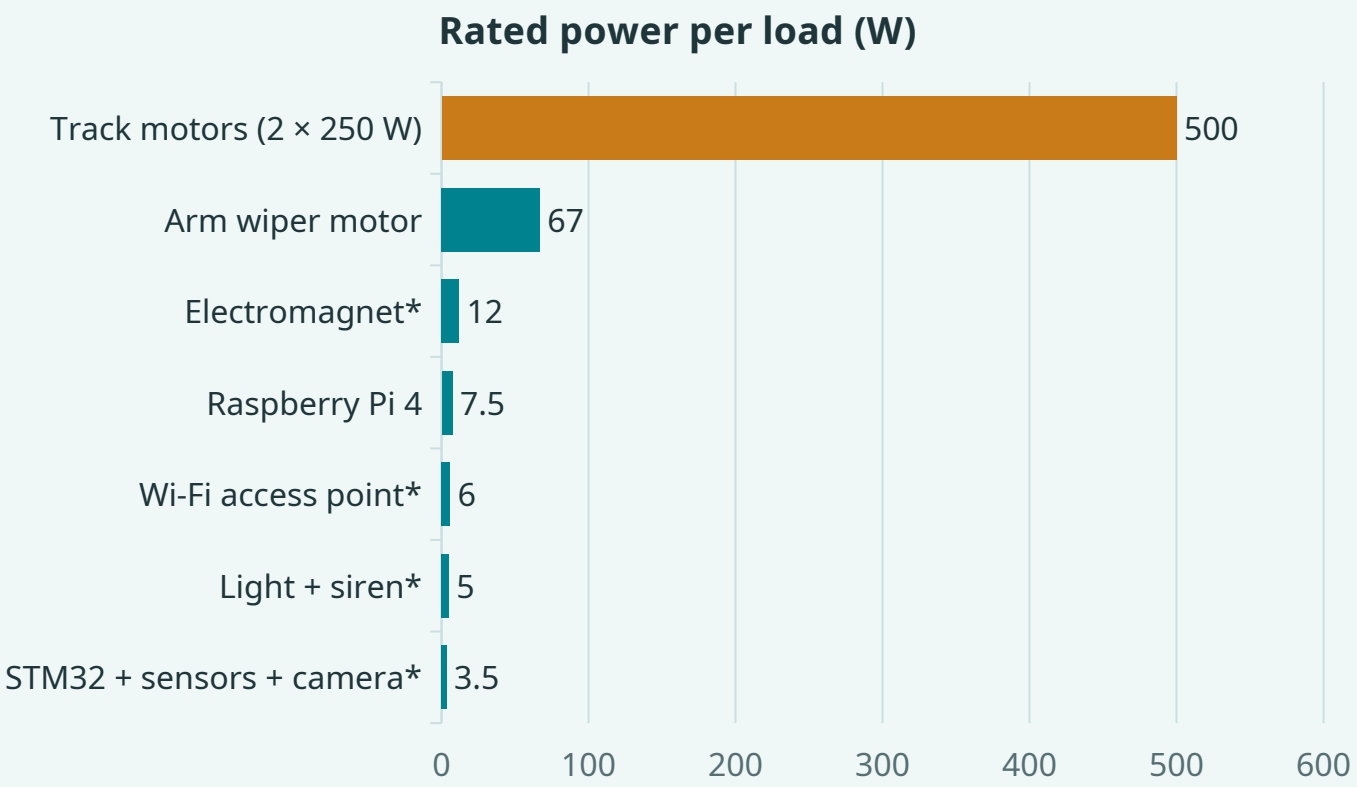
ROS 2 bridge
Docker

GUI + vision

ELECTRONICS

Power budget

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Motor values from datasheets (wiper: 12 V × 5.6 A). *Estimates, to be measured.

≈ 600 W

Peak, all loads at rating

≈ 230 W

Match average (drive at ~40 %)

≈ 75 Wh

Energy for one 20 min run

≈ 3 runs

From ≈ 222 Wh on board**

**Assumes the two 10 Ah packs form one 22.2 V (6S) bus. Peak ≈ 27 A, well inside the packs' discharge rating.

CONTROL SYSTEM

Stopping is layered

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Laptop → robot

1 — GUI focus lost
Drive zeroed, speed lock cleared

→

2 — GUI refresh
No UI update for 200 ms → zero

→

3 — ROS bridge
Drops commands > 200 ms old, publishes one zero

→

4 — Raspberry Pi
Own motion timeout

→

5 — STM32 watchdog
250 ms without valid frame → PWM 0

→

6 — Hardware E-stop
Cuts power to the loads

Every stage fails safe on its own. A lost link, hung GUI or crashed process stops the robot, and reconnecting never resumes motion: the operator must press a key again.

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03

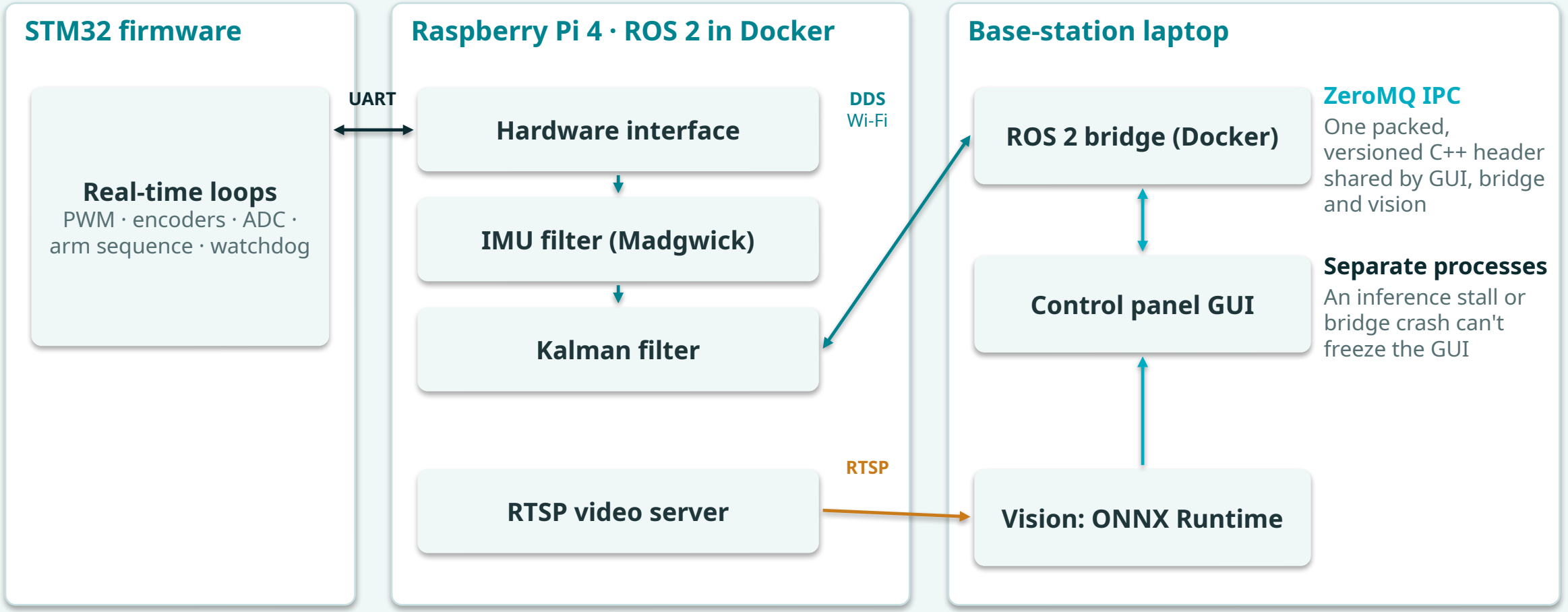
Software Stack

- Architecture: STM32, Pi and laptop
- Mapping and localization
- Mine detection and classification
- Control station GUI
- Navigation plan and reliability



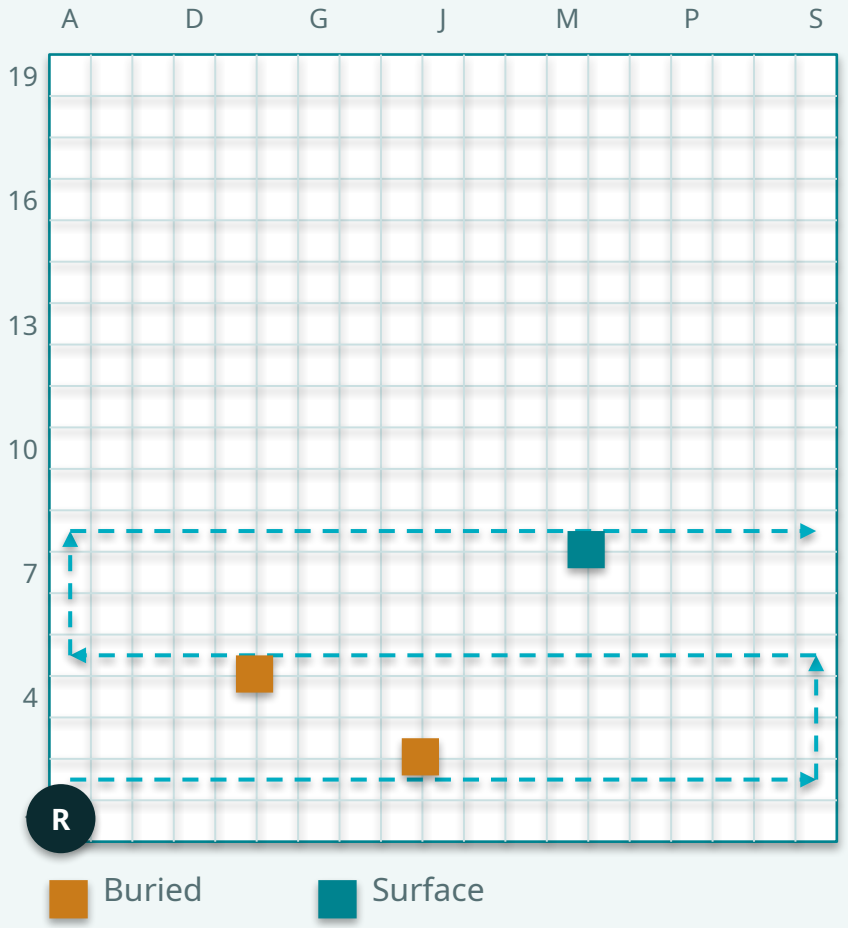
Architecture: three tiers, one contract

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SOFTWARE

Mapping and localization



1 Localize

Kalman filter fuses track odometry (STM32 encoders) with the MPU-9250 heading. No GPS.

2 Re-anchor

At a known corner the operator resets x, y and snaps the heading to the nearest 90°.

3 Place the mine

Robot pose + detector offset (dx, dy), rotated by heading θ:

$$x_m = x + dx \cdot \cos\theta - dy \cdot \sin\theta$$
$$y_m = y + dx \cdot \sin\theta + dy \cdot \cos\theta$$

4 Merge and persist

Hits within a merge radius are one mine; records survive a restart. The map is generated automatically.

SOFTWARE

Mine detection and classification

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Metal detector
rising edge

↓

Camera → ONNX
fresh vision result?

↓

Light + siren
on every metal hit

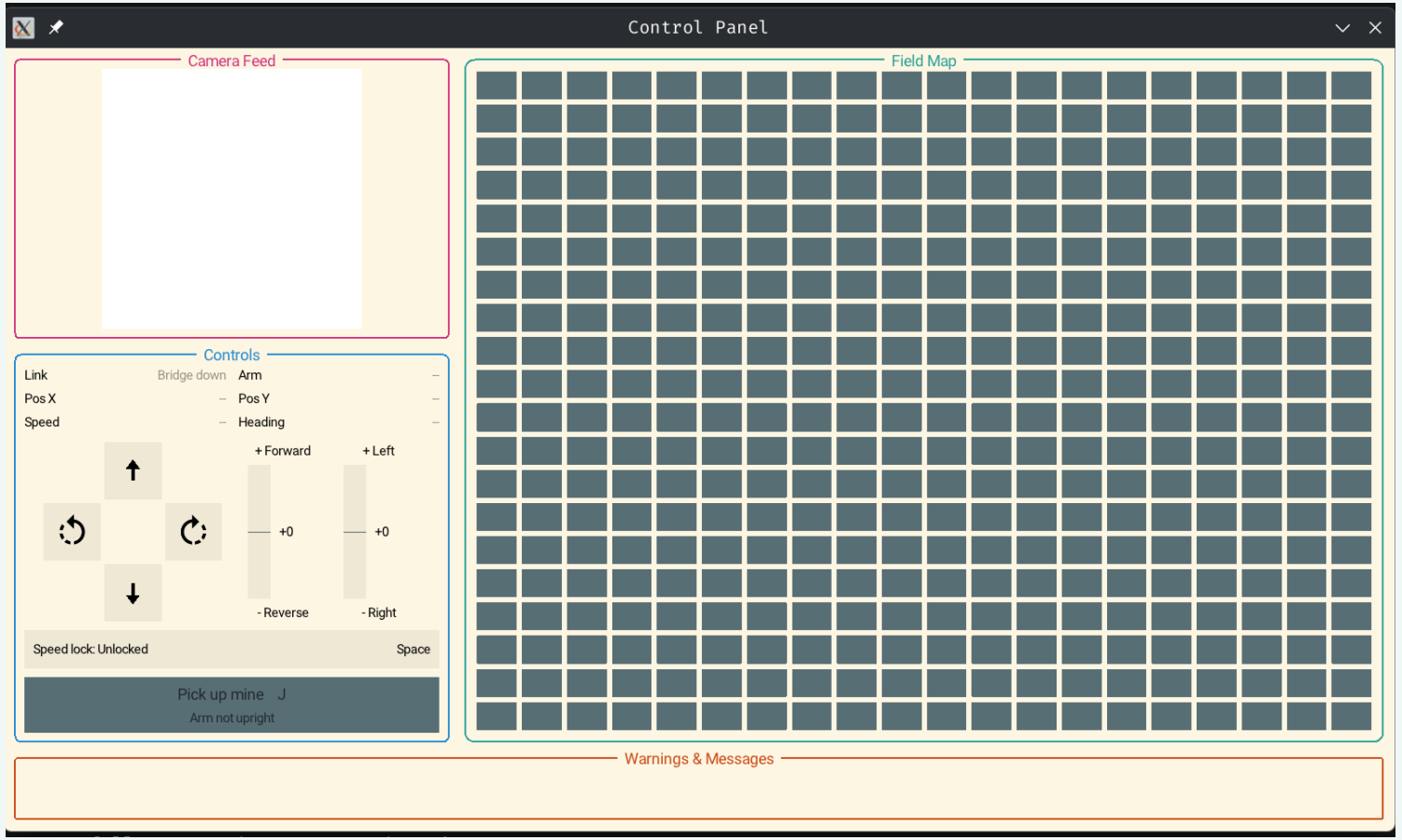
Metal + camera sees a box → **SURFACE** → Mapped, then the autonomous pickup sequence collects it

Metal + camera sees nothing → **BURIED** → Mapped at its position; the arm is not used

Vision alone never creates a mine: the metal detector must confirm it.

SOFTWARE

Control station GUI



1. **Camera feed**

   Live RTSP video from the robot

2. **Field map**

   Mines placed automatically as they are found

3. **Drive controls**

   Arrow keys with smooth ramp; Space locks speed

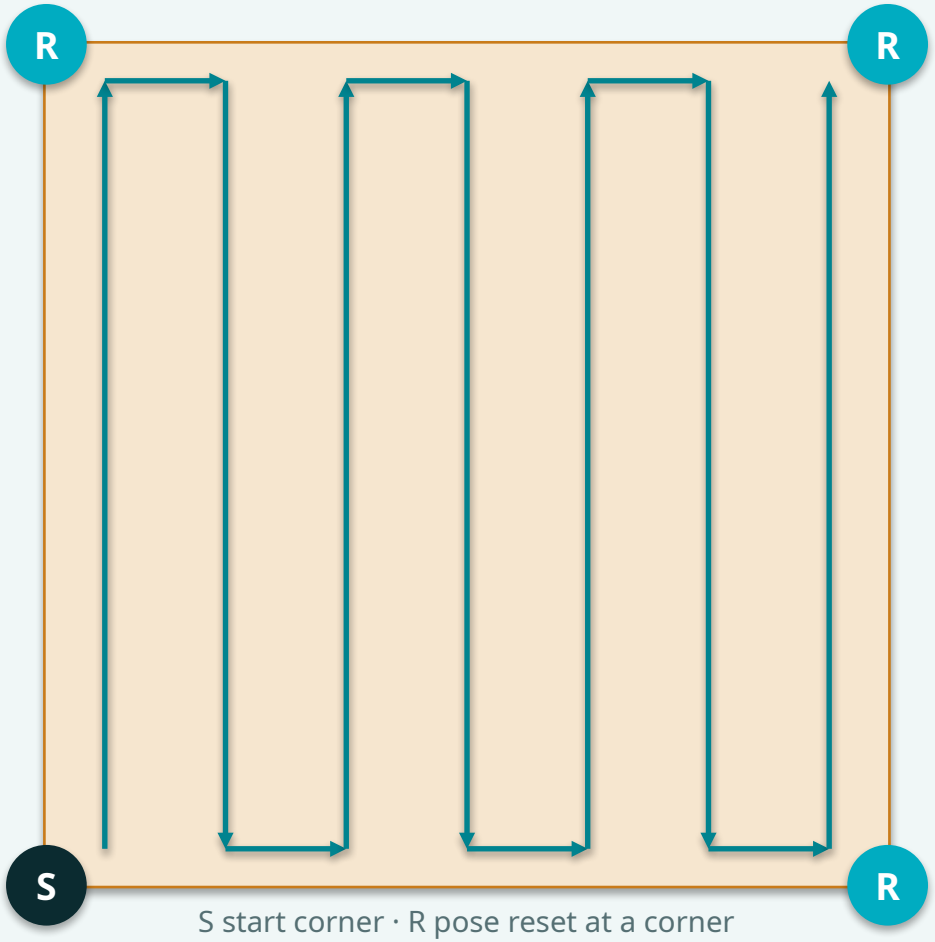
4. **Pick up mine (J)**

   Starts the autonomous pickup; disabled until the arm is upright

5. **Link health**

   Stale data greyed out, never shown as zero

Arena navigation plan



S start corner · R pose reset at a corner

1. **Start in a corner**

   Pose reset to the known corner before the first lane.

2. **Lawnmower sweep**

   Parallel lanes, spacing = detector width minus overlap, so no strip of the field is missed.

3. **Stop, signal, classify**

   Light and siren fire on every metal hit; surface mines are picked up on the spot.

4. **Re-anchor at every corner**

   Each lane end near a corner resets drift.

5. **Budget the 20 minutes**

   Sweep time ≈ (19 m / lane spacing) × 19 m / speed, planned to cover the whole field.

SOFTWARE

Built to be reliable

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1 One contract

A single versioned C++ header defines every message the GUI, bridge and vision process exchange.

2 Validate everything

Wrong size, version, flag, range or non-finite float: the message is dropped.

3 Dockerized robot stack

Host networking for DDS, devices found at boot, config bind-mounted so we can tune on site without rebuilding.

4 Tested

Unit and integration tests cover drive ramping, pickup gating, message decoding and the ZeroMQ link.

TEAM

Team, budget and next steps

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Watad Ejust

Academia category

Mechanical · electronics · software

Budget-minded build

Wheelchair drive motors and a car wiper motor bought from the local market; an open-source software stack.



After the competition

- Autonomous navigation: coverage planning on top of the existing localization
- Onboard vision on a Jetson; the GUI protocol stays the same

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Thank You

